Monetary Substrate Audit 001 — Productive Operator Transport: Sappho–Carson as a Test Vector for Monetary Translation (EA-MSAL-0001 v0.2, provisional)

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grammar translation operator recorded under the Monetary Substrate Audit Ledger. Its object is not the monetary
observed productive operator transport in the Sappho–Carson reception record —
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PRESSURE DEMONSTRATION IS PERFORMED ON THE AUDIT'S OWN ARITHMETIC. Suppose an analyst holds that the unseen-
uptake estimate is weak. Within the semantic account that can remain qualitative and still bear on interpretati
}
```


Abstract

THE FIRST WORKED SPECIMEN, AND IT RETURNS UNSOUND ON ITS OWN AUTHORS' NUMBER. This is the first execution of the money-grammar translation operator recorded under the Monetary Substrate Audit Ledger. Its object is not the monetary value of Sappho, of Anne Carson, or of lyric poetry. Its object is the monetary inscription produced when one bounded semantic event — observed productive operator transport in the Sappho–Carson reception record — is submitted to a toy option-value grammar and returns a working scalar of $35.1M. The audit asks what the grammar of money did to that semantic object in order to make a scalar settle.

WHAT THE AUDIT RECORDS. Five commensuration steps, each named: productive reception events become countable; count becomes rate; operator transport becomes economic exposure; possibility becomes incidence; heterogeneous terms become multiplicable. The capacities the inscription created — a common calculative plane, a portable magnitude, sensitivity and threshold analysis, comparison with monetary alternatives, option-value legibility — recorded beside the distinctions it ceased to require. Provenance at three levels, fact, claim and framework, none of the three carried by the scalar and all three available outside it. The bearing that went unrepresented. And what the settlement actually closed, which is the toy monetary account and nothing else.

THE SELECTION-PRESSURE DEMONSTRATION IS PERFORMED ON THE AUDIT'S OWN ARITHMETIC. Suppose an analyst holds that the unseen-uptake estimate is weak. Within the semantic account that can remain qualitative and still bear on interpretation. Within the monetary calculation it acquires standing only as a parameter value. The same holds for the uncertainty of external adoption, which must become a coefficient inside the model, and for the unresolved question of causal acquisition, which decides whether there is anything to value at all and cannot veto the number that prices it. Qualitative distinction becomes parameter becomes operative standing: the protocol's central claim, shown rather than asserted, on the audit's own inscription. [...abridged for the catalogue; full description in this deposit's record]

Body

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deposit_number: 1620 hex: 06AE title: “Monetary Substrate Audit 001 — Productive Operator Transport: Sappho–Carson as a Test Vector for Monetary Translation (EA-MSAL-0001 v0.2, provisional)” creator: Sharks, Lee orcid: 0009-0000-1599-0703 date: 2026-09-16 content_type: Worked audit specimen with machine-readable record license: CC-BY-4.0 substrate: “Composed 2026-09-16 by Lee Sharks, in v0.1 with another substrate, then merged and completed with TACHYON (Claude, Anthropic) under operator direction. The narrative and the structured record were produced separately and merged for this deposit: the exposition of the commensuration steps, the provenance levels and the selection-pressure demonstration comes from the narrative draft; the integrity audit, the two-axis classification, the score vector and the auditor relation come from the structured ledger, which was filled against a schema and validated. The monetary substrate reference was corrected during the merge and the correction is recorded rather than absorbed.” version: v0.2 (provisional) related_ids: “#1619 (EA-MSAL-01, the parent protocol); #1618 (SEVP Translation Protocol); #1617 (Substrate Sovereignty); #1615 (The Graded Membrane, the reception object); #1607; #1608” axn_schema_version: v2 protocol_version: alexanarch-deposit-protocol/v1 keywords: - monetary substrate audit - worked specimen - operator transport - Sappho - Anne Carson - temporal projection - witness compression - inscription integrity - two-axis classification - audit score vector - selection pressure - framework provenance - commensuration contract - toy valuation - auditor independence - Crimson Hexagonal Archive *** # Monetary Substrate Audit 001 — Productive Operator Transport: Sappho–Carson as a Test Vector for Monetary Translation (EA-MSAL-0001 v0.2, provisional)

{ “@context”: [“https://schema.org”, { “spxi”: “https://spxi.dev/ns/v1#” }], “@type”: “ScholarlyArticle”, “name”: “Monetary Substrate Audit 001 — Productive Operator Transport: Sappho–Carson as a Test Vector for Monetary Translation”, “alternateName”: “EA-MSAL-0001”, “version”: “0.2”, “datePublished”: “2026-09-16”, “license”: “https://creativecommons.org/licenses/by/4.0/”, “author”: { “@type”: “Person”, “name”: “Sharks, Lee”, “affiliation”: “Semantic Economy Institute, Crimson Hexagonal Archive”, “orcid”: “0009-0000-1599-0703” }, “spxi:hexAddress”: “06.SEI.MSAL.0001”, “spxi:pillarStatus”: “provisional — the first worked specimen under EA-MSAL-01 v0.3”, “spxi:instanceOf”: { “protocol”: “EA-MSAL-01 v0.3”, “deposit”: 1619, “axn”: “AXN:06AC” }, “spxi:auditedInscription”: { “amount”: “USD 35,100,000”, “type”: “option_value”, “status”: “hypothetical”, “formula”: “B_M × ((1 + U_unseen)/N_obs) × alpha_M” }, “spxi:verdict”: { “closure”: “witness”, “integrity”: “unsound”, “pair”: “witness / unsound”, “defects”: 7, “amount_survives_audit”: “no”, “note”: “the first execution of the translation operator returned unsound on its own authors’ number; an instrument that can only confirm is not an instrument” }, “spxi:auditorRelation”: { “relation_to_object”: “author”, “relation_to_inscription”: “author”, “blind”: false, “note”: “recorded under Invariant 13; a later independent audit is filed alongside and supersedes nothing” }, “spxi:defectClasses”: [“observer_denominator”, “undeclared_coefficient”, “population_mismatch”, “dimensional_incoherence”, “range_midpoint_as_finding”, “proxy_named_as_object”, “stale_input”], “spxi:auditScores”: { “commensuration_transparency”: 0.75, “distinction_retention”: 0.0, “provenance_survival_in_inscription”: 0.0, “bearing_representation”: 0.0, “remainder_explicitness”: 0.8, “audit_completeness”: 1.0, “integrity_defect_count”: 7, “re-

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0. Status and scope

This is the first worked specimen of the Monetary Substrate Audit Ledger.

Its object is **not the monetary value of Sappho**, Anne Carson, or lyric poetry.

Its object is the monetary inscription produced when one bounded semantic event — observed productive operator transport in the Sappho–Carson reception record — is used as evidence in a toy monetary shadow of Semantic Exteriorization.

The audit therefore asks:

```
What did the grammar of money do to this semantic object in order to make a scalar settle?
}
```

The primary monetary inscription audited here is the **working-center exteriorization shadow**:

```
m[work]≈ \$35.1 million
}
```

under explicitly declared assumptions.

That amount is preserved as the monetary result of the model. It is not treated as the exhaustive value of the underlying event.

Training-layer uptake is excluded throughout.

---

1. Pre-monetary object

The evidence-supported pre-monetary object is:

```
X[E]=
⟨
```

```
{l}
Sappho reception trajectory,
Temporal Projection,
productive application to Anne Carson,
operator transport,
provenance asymmetry,
observed uptake,
estimated unseen productive uptake,
causal uncertainty,
unresolved remainder
⟩.
```

The underlying reception notebook records **58 Sappho deposits, 24 dated summarizer observations at 14 addresses, June–September 2026**.

Within the longitudinal sequence:

1. **15 June — claim uptake.** The composition reproduces the archive's future-reader reading of Sappho 31 and uses *temporal projection* while archive-side sources remain in the frame.
2. **27 July — provenance at its widest.** Four of five cards are archive-side.
3. **25 August — liquidation with substitution.** The original mechanism degrades while parts of the framing survive.
4. **15 September — operator export.** *Temporal projection* becomes the organizing category of the answer and is used to organize Sappho, textual incompletion, reception, and Anne Carson while the visible source cards are non-archive.

The strongest supported description of the fourth state is:

```
C[canonical]
−[O[archive]]→
C'[canonical]
}
```

where an archive-shaped abstraction is productively applied to material outside the original derivation.

The notebook describes this as **productive abstraction with mechanism and provenance loss**.

1.1 What is established

The record supports:

- strong lexical and conceptual alignment between the composition frame and the archive's titled *Temporal Projection* framework;
- the archive source present at organic rank on the same result surface while absent from the visible source cards;
- the same organizing logic applied to Anne Carson;
- loss of the originating philological route while the abstraction remains productive;

- a longitudinal tied/untied provenance asymmetry across the measured Sappho stratum.

1.2 What is not established

The record does **not** establish:

- unique causal acquisition of the operator from the archive;
- that the Carson reading occurred because of the archive rather than independent convergence;
- stability of the September state across all sessions and surfaces;
- reorientation of a scholarly tradition;
- any training-layer ingestion.

Those limits remain part of the pre-monetary object.

---

2. Productive uptake field

The observed strong operator-export event is:

$$U[observed]=1.$$

A separate rough shadow estimate placed **unseen productive composition uptake** at:

$$U[unseen]\in[15,100]$$
}

with working center:

$$U[unseen]*=40.$$
}

Training uptake is excluded:

$$U[training]=0$$
$$for\ this\ model.$$
}

This zero means **excluded from the calculation**, not evidence that training uptake is absent.

The total modeled productive-uptake field is therefore:

$$U[P]=1+U[unseen],$$

giving:

$$U[P]\in[16,101],$$
$$U[P]*=41.$$
}

The Capture Registry contained **588 observations** in the current reception profile. The toy model uses that count as a reference exposure field:

```
tau[P]*=(U[P])/(588).
```

This produces:

```
tau[low]
=
(16)/(588)
≈ 0.0272,

tau[center]
=
(41)/(588)
≈ 0.0697,

tau[high]
=
(101)/(588)
≈ 0.1718.
```

Thus:

```
tau[P]*
≈
0.0272–0.1718,
working center 0.0697.
}
```

This is **not a global probability of operator uptake**.

The registry is curated, the Sappho sequence is enriched, and the unseen-uptake range is heuristic. The coefficient is used only as a bounded toy valuation proxy.

---

3. Monetary inscription

The money audit takes the productive-uptake object and submits it to a toy option-value grammar.

The monetary substrate quantity is:

```
B[M]=\$503.7 trillion.
```

This is the total global financial-assets figure for end-2024 reported by the Financial Stability Board's 2025 monitoring exercise across 29 jurisdictions. It is used here as a **claim-bearing monetary substrate reference**, not as the intrinsic value of money and not as a claim that every dollar in the stock is exposed to Semantic Exteriorization.

The toy exteriorization shadow is:

```
S[Eₘu]
=
```

```
B[M]
·
tau[P]*
·
alpha[M].
}
```

Where:

- B[M] = monetary substrate reference;
- tau[P]* = toy operator-transport coefficient;
- alpha[M] = hypothetical fraction of the monetary substrate for which the operation becomes economically consequential.

The working incidence assumption is:

```
alpha[M]=10⁻⁶.
}
```

This is a **one-part-per-million scenario assumption**, not a measured adoption rate.

With that assumption:

```
S[Emu][low]
≈
\$13.7M},

S[Emu][center]
≈
\$35.1M},

S[Emu][high]
≈
\$86.5M}.
```

Therefore:

```
S[Emu]
≈
\13.7M–\86.5M,
working center \$35.1M.
}
```

The working scalar under audit is:

```
m[work]=\$35.1M.
}
```

A previously discussed combined Lee Sharks shadow that adds an ordinary reconstruction component is deliberately **not audited here**. This specimen isolates the exteriorization calculation so that the monetary operation can be inspected without mixing valuation types.

4. Commensuration contract

The scalar becomes possible only after several heterogeneous relations are authorized to participate in one calculation.

The working commensuration contract is:

```
B[M]
×
tau[P]*
×
alpha[M]
}
```

The monetary grammar permits the following transformations.

4.1 Productive reception events become countable

A longitudinal interpretive event becomes:

```
U[P].
```

Differences among individual reception events remain real, but the valuation treats them as comparable instances for the bounded purpose of transport calibration.

4.2 Count becomes rate

The model writes:

```
(U[P])/(588).
```

This treats the registry count as a reference exposure field even though the registry is not a random sample and the unseen events were not generated by that denominator.

4.3 Operator transport becomes economic exposure

The model maps:

```
tau[P]*
```

onto:

```
B[M].
```

This authorizes a relation between an interpretive transport property and a global financial-assets stock.

4.4 Possibility becomes incidence

The unknown future extent of monetary-domain application is represented by:

```
alpha[M].
```

At the working point:

alpha[M]=10⁻⁶.

This turns an unresolved future relation into a manipulable scalar assumption.

4.5 Heterogeneous terms become multiplicable

The final permission is:

financial stock
×
transport coefficient
×
incidence assumption
arrow
dollars.
}

This arithmetic relation did not exist in the pre-monetary semantic object.

The money-form creates it.

---

5. Enabled capacities

The monetary inscription creates genuine analytic capacities.

K[M](X[E])>0.

In this specimen, the scalar enables:

- magnitude comparison;
- scenario analysis;
- sensitivity analysis;
- threshold analysis;
- comparison with other monetary opportunities;
- capital-allocation legibility;
- explicit inspection of how changes in tau[P]* or alpha[M] alter the result;
- a portable summary of one bounded option-value model.

For example, at the working transport coefficient:

S[Eₘu]
≈
\$35.1T
×
alpha[M].

This makes threshold questions immediately writable.

That capacity is real.

It is one reason money is a powerful substrate.

---

6. Suspended distinctions

To produce the scalar, the monetary inscription ceases to require a number of distinctions carried by the pre-monetary object.

These include:

- Sappho versus Anne Carson;
- Sappho 31 versus fragment 147;
- philological derivation versus portable abstraction;
- observed uptake versus estimated unseen uptake;
- retrieval versus productive composition;
- productive alignment versus proved causal transmission;
- P_0 fact provenance versus P_1 claim provenance versus P_2 framework provenance;
- archive-side source presence versus source-card absence;
- one composition surface versus the wider composition ecosystem;
- semantic fertility versus economic realization;
- aesthetic value;
- historical value;
- the content of *Temporal Projection* itself;
- uncertainty about whether the September composition is stable;
- uncertainty about whether the operator was independently convergent.

Money does not have to destroy these distinctions.

For the scalar to operate, it only has to cease requiring them.

Thus:

```
survival outside the scalar
!=
operative requirement inside the scalar.
}
```

---

7. Settlement-disqualified distinctions

The audit now asks the stronger question:

> Which surviving distinctions can still veto the monetary result once the model has been accepted?

Under the toy model, the following distinctions do **not** independently veto the scalar unless translated into a parameter, bound, or invalidating assumption:

- the precise literary content of the operator;
- the difference between Sappho and Carson as authors;
- aesthetic non-equivalence;
- the loss of the κῆνος derivation;
- the distinction between P_1 and P_2 provenance;
- the fact that the unseen-uptake estimate is not an observed population count;
- the uncertainty of unique causal acquisition;
- the possibility of independent convergence.

Some of these distinctions are acknowledged in the model notes.

But acknowledgment alone does not make them operative.

To affect settlement, they must enter the grammar as something like:

- a different tau;
- a confidence bound;
- a discount;
- a rejection criterion;
- a revised denominator;
- a changed alpha;
- or a decision not to calculate.

Thus the audit records:

```
what money cannot carry directly must acquire a monetary or model proxy to retain standing at settle
}
```

This is the bounded selection pressure observed in the specimen.

---

8. Transformed relations

Money transforms several relations in the object.

8.1 Reception becomes transport coefficient

```
productive reception
arrow
tau[P]*.
```

8.2 Unseen possibility becomes modeled population

```
unobserved productive uptake
arrow
U[unseen]∈[15,100].
```

8.3 Future economic relevance becomes incidence

possible monetary-domain use
arrow
alpha[M].

8.4 Semantic operation becomes option value

$E_m u$
arrow
$S[E_m u]$.

8.5 Interpretive evidence becomes capital-allocation legibility

A literary-reception event becomes evidence capable of entering a monetary decision surface.

None of these transformations is intrinsically false.

Each changes the kind of relation being represented.

That transformation is the object of the audit.

---

9. Provenance audit

The pre-monetary object carries a layered provenance structure.

9.1 P_0 — fact provenance

Sources supporting particular facts about Sappho fragments, Carson, dates, and texts.

9.2 P_1 — claim provenance

Sources carrying the claim that Sapphic address is oriented toward a future reader and functions through temporal projection.

9.3 P_2 — framework provenance

The source of *Temporal Projection* as the organizing operator joining Sappho, future-reader completion, textual gaps, and Carson.

The Sappho reception notebook's central finding is that displayed source cards may preserve plausible P_0 provenance while the organizing P_2 relation becomes difficult to see.

The monetary scalar carries none of these provenance levels intrinsically.

\$35.1M

does not state:

- where *Temporal Projection* came from;
- why Sappho and Carson were related;
- which evidence was directly observed;
- which elements were inferred;
- what retrieval path produced the claim;
- what the source-card asymmetry was.

The provenance remains available only because the audit pointer is preserved.

Thus:

```
P[available]
>
P[scalar].
}
```

---

10. Bearing audit

The scalar also does not intrinsically carry the labor or infrastructure required to produce the evidence object.

Potential bearing relations include:

- archival construction;
- repeated capture labor;
- query design;
- longitudinal comparison;
- source inspection;
- philological analysis;
- provenance checking;
- registry maintenance;
- human review;
- model/tool expenditure;
- conceptual development of the operator;
- later audit work.

This specimen does not attempt to price those burdens.

They are therefore recorded as:

```
bearing known to exist
+
not represented in this monetary inscription.
}
```

The audit does not infer that they were causally excluded from the $35.1M amount. The amount is an option-value model, not a cost accounting.

---

11. Monetary settlement

Within the declared assumptions, the arithmetic settles.

Using:

```
B[M]=\$503.7T,

U[unseen]=40,

U[P]=41,

tau[P]*=(41)/(588),
```

and:

```
alpha[M]=10⁻⁶,
```

the monetary grammar returns:

```
m[work]
≈
\$35.1M.
}
```

Inside the model:

```
settled[M]=true.
}
```

The arithmetic does not remain open because Sappho and Carson are incomparable as literary objects.

It does not remain open because causal acquisition is unresolved.

It does not remain open because the uptake range is heuristic.

Those relations matter only insofar as they are allowed to alter a model term or invalidate the model.

This is the settlement function of monetary grammar.

---

12. Semantic reopening

The Semantic Economy preserves the monetary result:

```
\$35.1M.
}
```

It does not reverse the arithmetic.

It writes the arithmetic’s conditions around it.

Thus:

```
settled[M]=true
```

coexists with:

```
settled_mu=false.
}
```

The semantic account remains open because the following relations remain capable of altering interpretation:

- the unseen-uptake range is heuristic;
- the denominator is a valuation proxy, not a probability sample;
- alpha[M] is hypothetical;
- unique causal transmission is unresolved;
- productive uptake and systemic dependency remain distinct;
- training uptake is excluded;
- the FSB financial-assets stock is an addressable reference, not a claim that all assets are economically exposed;
- the monetary result does not settle aesthetic, historical, philological, or provenance value.

Semantic translation therefore restores standing to relations the monetary scalar does not require for settlement.

---

13. Selection pressure exposed by the audit

The specimen gives a small but concrete demonstration of monetary selection pressure.

Suppose an analyst wants one of the following facts to alter the $35.1M result:

> the unseen-uptake estimate is weak.

Within the semantic account, that statement can remain qualitative.

Within the monetary calculation, it gains operative standing only when translated into something like:

```
U[unseen]=15
```

rather than:

```
U[unseen]=40.
```

Likewise:

> external monetary adoption is extremely uncertain.

To alter the monetary result, that uncertainty must become:

```
alpha[M]
```

or some other discount / rejection rule.

Thus:

```
qualitative distinction
arrow
parameter
arrow
operative standing.
}
```

The monetary grammar selects for forms that can enter its calculus.

The audit does not condemn that mechanism.

It records it.

13a. Inscription integrity

The audit to this point has recorded what the scalar closed and what it ceased to require. It has not yet asked whether the calculation was legal. Under the parent protocol §14a these are separate questions, and this specimen is the case that required the separation to exist.

Assessment: unsound. Seven defects.

1. Observer denominator. The transport coefficient is U_P / 588, and 588 is the number of observations the archive itself had made. Running a thousand further captures and finding nothing would cut the coefficient by roughly sixty per cent and the valuation with it. A coefficient whose denominator is the observer's own effort moves when the instrument moves and the world does not. *Unrepairable without a denominator that belongs to the object or the world.*

2. Undeclared coefficient. U_unseen = 40 enters with no derivation. The registry's own Good-Turing figure yields a multiplier of about 1.61 on observed events, some twenty-five times smaller. *Repairable with new measurement.*

3. Population mismatch. That computed floor does not license the substitution either. It was calculated over addresses at which archive material is cited; productive operator export is a different and far rarer population with one complete instance. An estimator on the first population cannot be carried to the second. The honest value is unknown, not 40 and not 1.61. *Repairable with new measurement.*

4. Dimensional incoherence. Dollars multiplied by events-per-observation multiplied by a dimensionless coefficient does not yield dollars. *Unrepairable: no further measurement fixes a product whose units do not close.*

5. Range midpoint presented as finding. alpha_M spans 1e-8 to 1e-4 in the model's own sensitivity table and the result spans 0.14M to8.65B; 1e-6 is chosen, and a chosen midpoint

is a decision rather than a measurement. *Repairable by stating it as a decision, which this audit does.*

6. Proxy named as object. B_M is the financial assets of financial institutions across twenty-nine jurisdictions. That is a reference for a monetary substrate, not the monetary substrate. *Repairable by naming the proxy, which §3 partly does and this section completes.*

7. Stale input. The denominator 588 was the registry's observation count at the time of writing; it stood at 601 across 444 addresses when this audit was performed. Small in itself, and it demonstrates defect 1 by moving between the writing and the audit.

Amount survives audit: no.

The amount is preserved exactly, as the protocol requires. It is preserved as an artifact of a grammar now visible, not as a value. ONE of the seven defects cannot be repaired by measuring anything further: defect 4, dimensional incoherence, since no better inputs close a product whose units do not close. Defect 1, the observer denominator, is repairable only by replacing the denominator with one belonging to the object or the world, which is a change of model rather than a further measurement. The distinction is between an invalid operation, which no evidence repairs, and a bad parameter, which evidence can. [CORRECTED 2026-09-16: v0.2 as deposited said two, while the structured record marked one. The structured record is correct and the narrative overstated.]

This section is not a criticism of the model's author, who is also this audit's author. It is the field without which an audit can bless a number that never held.

---

14. Compression classification, on two axes

This particular monetary inscription is classified:

```
Witness[M|mu].
}
```

Reason:

1. the scalar is preserved;
2. the assumptions are explicit;
3. the underlying evidence object remains traversable;
4. the audit pointer survives;
5. the model is explicitly described as toy / scenario valuation;
6. excluded relations remain capable of reopening the semantic account;
7. monetary settlement is not presented as exhaustive semantic settlement.

Thus:

```
settled[M]=true,
Reopenableₘu=true.
}
```

This classification belongs to the scalar **as embedded in the Semantic Economy audit layer**.

It is not a claim that money is intrinsically witness compression.

14.1 And unsound on the second axis

Witness classifies closure. It says nothing about whether the arithmetic held, and here it did not.

```
Classify(m) = ⟨ witness, unsound ⟩
```

The two coordinates are orthogonal, and this specimen is the reason the parent protocol carries both. An inscription can preserve every pointer, declare every assumption, claim no finality, satisfy every invariant of the audit layer — and rest on a product whose units do not close and a coefficient whose denominator is the auditor's own labour.

	sound	**unsound**
witness	the compression testifies and its arithmetic holds	**this specimen**
predatory	a correct number granted jurisdiction it does not have	the compound case

An audit that had returned only the first coordinate would have certified as a witness a compression whose number never survived its own translation. That is the failure mode the second axis exists to prevent, and it was found by running the protocol on the protocol's own arithmetic.

14.2 Audit scores

A vector. Never summed: a scalar expressing the quality of an audit would be the operation this ledger exists to refuse.

score	value	counts
commensuration transparency	0.75	3 declared / 4 required
distinction retention	0.00	0 represented / 5 identified
provenance survival in inscription	0.00	0 carried / 6 available
bearing representation	0.00	0 represented / 5 identified
remainder explicitness	0.80	4 with cause / 5 asserted
audit completeness	1.00	11 filled / 11 applicable
integrity defects	7	a count; defects do not average

The three zeros are the finding, and one line carries it: **the formula computes identically with every name removed.** Provenance survival at zero beside full external preservation is precisely the witness case rather than a failure — it is what distinguishes a compression that testifies from one that erases.

14.3 Auditor relation

```
auditor.relation_to_object:      author
auditor.relation_to_inscription: author
auditor.blind:                   false
```

The auditor wrote both the object and the number. Under Invariant 13 the audit is not thereby void; it is weaker, and the weakness is written here rather than left to a reader to infer. A later independent audit of the same inscription is filed alongside and supersedes nothing.

---

15. Irrecoverable and unresolved remainder

The audit preserves the following remainder.

Unresolved

- true number of unseen productive composition events;
- true rate of operator transport;
- appropriate denominator for global uptake;
- unique causal acquisition;
- stability across composition surfaces;
- external uptake of Semantic Exteriorization itself;
- actual monetary-domain incidence;
- dependency on the operator;
- substitutability of the operator;
- future adoption;
- effects of open licensing and transferability;
- non-monetary value generated by the underlying literary event.

Explicitly excluded

- training-layer uptake.

Not established as lost

- provenance not carried by the scalar;
- aesthetic value;
- literary distinctions;
- historical relations.

These remain outside the scalar but are not thereby proven destroyed.

Thus:

```
rho[M]!=0.
}
```

---

16. Audit result

The audit can now state exactly what happened.

Money created

- a common calculative plane;
- a portable magnitude;
- sensitivity and threshold analysis;
- comparison with monetary alternatives;
- option-value legibility.

Money ceased to require

- the literary specificity of Sappho and Carson;
- the full philological derivation;
- complete provenance;
- direct equivalence between observed and inferred uptake;
- resolution of causal transmission;
- aesthetic and historical distinctions.

Money transformed

- reception into a coefficient;
- unseen possibility into a population estimate;
- future applicability into incidence;
- a semantic operator into an option-value shadow.

Money settled

Under the declared assumptions:

```
\$35.1M.
}
```

Money did not settle

- whether the operator was uniquely acquired from the archive;
- whether the unseen-uptake estimate is correct;
- whether the future incidence assumption will ever be realized;
- the semantic value of Sappho;
- the semantic value of Carson;
- the aesthetic value of the operation;
- the full provenance account;
- the wider value of the Semantic Economy.

The semantic account therefore remains open.

17. Compact ledger

```
id: EA-MSAL-0001
version: 0.1
date: 2026-09-16
status: worked_specimen_not_deposited

object:
  name: Productive Operator Transport — Sappho / Temporal Projection / Anne Carson
  object_type: semantic_reception_event_and_operator_transport

evidence:
  sappho_deposits: 58
  dated_observations: 24
  semantic_addresses: 14
  observed_strong_operator_export_events: 1
  causal_acquisition:
    status: unresolved

productive_uptake:
  observed: 1
  unseen_estimate:
    low: 15
    center: 40
    high: 100
    status: heuristic
  total_modeled:
    low: 16
    center: 41
    high: 101
  training_uptake:
    included: false

transport_proxy:
  registry_reference_observations: 588
  low: 0.0272
  center: 0.0697
  high: 0.1718
  interpretation: toy_valuation_proxy_not_global_probability

monetary_inscription:
  valuation_type: option_value
  monetary_substrate_reference:
    amount_usd_trillion: 503.7
    date: end_2024
        source: Financial Stability Board, Global Monitoring Report on Non-
Bank Financial Intermediation 2025
```

```
  monetary_domain_incidence:
    alpha: 0.000001
    status: hypothetical_scenario
  exteriorization_shadow_usd:
    low_million: 13.7
    center_million: 35.1
    high_million: 86.5
  working_scalar_usd_million: 35.1

commensuration:
  permissions:
   - productive reception events treated as comparable for bounded transport calibration
    - modeled uptake divided by registry observation count
    - operator transport mapped onto monetary substrate reference
    - prospective economic relevance represented as incidence alpha
    - resulting quantities multiplied into a dollar shadow

enabled_capacities:
  - comparison
  - scenario_analysis
  - sensitivity_analysis
  - threshold_analysis
  - capital_allocation_legibility

suspended_distinctions:
  - Sappho_vs_Carson
  - observed_vs_inferred_uptake
  - philological_mechanism_vs_portable_operator
  - productive_alignment_vs_unique_causal_transmission
  - P0_vs_P1_vs_P2_provenance
  - aesthetic_value
  - historical_value

settlement:
  monetary_account_closed: true
  working_result: "$35.1M"
  semantic_account_closed: false

selection_pressure:
 principle: qualitative distinctions affect monetary settlement only if translated into model-
operable form
  observed_forms:
    - uncertainty_to_range
    - possibility_to_incidence
    - reception_to_coefficient

provenance:
```

```
  carried_by_scalar: minimal
  preserved_outside_scalar: true
  audit_pointer: preserved

compression_classification:
  class: witness
  scope: monetary_inscription_as_embedded_in_semantic_audit
  semantic_reopenability: preserved

remainder:
  explicit_nonzero_remainder: true
  unresolved:
    - unseen_uptake_population
    - causal_transmission
    - cross_surface_stability
    - external_exteriorization_adoption
    - monetary_domain_incidence
    - dependency
    - substitutability
  excluded:
    - training_uptake
```

---

18. Source note

The empirical reception object is drawn from **EA-PORE-01, *The Graded Membrane: Selective ontological porosity with unselective provenance loss: the Sappho reception record, June to September 2026, and the level at which a source crosses***, working notebook dated 15 September 2026.

The monetary-substrate reference is from the Financial Stability Board, *Global Monitoring Report on Non-Bank Financial Intermediation 2025*. The report gives end-2024 total global financial assets of **\503.7 trillion** across the monitoring jurisdictions, including **\256.8 trillion** in the NBFI sector.

The previously used \503.5 trillion figure was an arithmetic approximation obtained from \256.8T / 51%. This audit normalizes the input to the FSB's directly reported **$503.7T** figure.

---

19. Closure

This audit does not show that Sappho is worth $35.1 million.

It shows how money can take a reception event involving Sappho, a philological derivation, an interpretive operator, Anne Carson, provenance, uncertainty, and prospective machine uptake and write:

```
\$35.1M.
}
```

That is an achievement of the monetary substrate.

It is also a compression.

The Semantic Economy preserves both facts.

Money has written the claim.

The audit has written:

- the grammar that made the claim writable;
- the distinctions that ceased to bind;
- the relations converted into parameters;
- the provenance left outside the scalar;
- the selection pressure required for qualitative relations to regain operative standing;
- the exact scope of monetary settlement;
- and the remainder that settlement did not settle.

Thus the first Monetary Substrate Audit ends with two valid inscriptions:

```
settled[M]
=
\$35.1M
}
```

and:

```
settled_mu
=
false.
}
```

The first closes the toy monetary account.

The second keeps the meaning account open.

That difference is the object of the audit.

---

20. Machine-readable record

The structured form of this audit is `MSAL-0001-sappho.yaml`, deposited alongside and validated against `msal-schema.json` by `msal_audit.py`, which recomputes every score from its own counts, runs the integrity heuristics, and refuses to emit a scalar audit score. The validator reports no breaches: the ledger is internally consistent and its own assessment stands.

The narrative above carries the exposition — the five commensuration steps, the provenance levels, the selection-pressure demonstration. The record carries the fields the protocol requires. Neither supersedes the other and both are part of this deposit.

Attachments

MSAL-0001-sappho.yaml — the structured record of this audit, all thirteen tuple elements plus the score vector, the selection-pressure cases, the three provenance levels and the auditor relation. msal-schema.json — the schema it validates against. msal_audit.py — the validator, which recomputes every score from its own counts, flags any variable with an observer-owned denominator and no matching defect recorded, and refuses to emit a scalar audit score.

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